

JEE MAIN REVISION MASTER: CAPACITOR

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 17 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Capacitor Combinations, Energy Stored & Charge Sharing	12 PYQs	Circuits & Energy
High	Parallel Plate Capacitance & Dielectric Slabs	9 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Capacitor Combinations, Energy Stored & Charge Sharing	$C_p = \sum C_i$, $\frac{1}{C_s} = \sum \frac{1}{C_i}$, $U = \frac{1}{2} C V^2$, $\Delta U_{\text{loss}} = \frac{C_1 C_2}{2(C_1 + C_2)} (V_1 - V_2)^2$
Parallel Plate Capacitance & Dielectric Slabs	$C = \frac{K \epsilon_0 A}{d}$, $E = \frac{V}{d} = \frac{\sigma}{\epsilon_0}$, $Q = C V$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Capacitor Combinations, Energy Stored & Charge Sharing (High Yield — 12 PYQs)

Core Principle: Series combination $(1 / C_s) = (1 / C_1) + (1 / C_2)$ (charge Q is same). Parallel combination $C_p = C_1 + C_2$ (V is same). Energy stored $U = (1 / 2) C V^2 = (Q^2 / 2C)$. Common potential on connecting two capacitors: $V_{\text{common}} = (C_1 V_1 + C_2 V_2 / C_1 + C_2)$.

■ **Caution / JEE Pitfall:** In series capacitors, voltage divides inversely to capacitance: $V_1 = V \left(\frac{C_2}{C_1 + C_2} \right)$. Smaller capacitor takes larger voltage!

■ **Speed Trick:** For n identical capacitors of capacitance C : $C_{\text{parallel}} / C_{\text{series}} = n^2$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "In the circuit shown in the figure, the potential difference across the $4\mu F$ capacitor is..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "A parallel plate capacitor has a uniform electric field ' E ' in the space between the plates. If the distance between the plates is ' d ...'" [Ans: C]
- [2024 Shift] Q#3 (MCQ): "A capacitor of capacitance $5\mu F$ is charged by a battery of emf $10 V$. At an instant of time, the potential difference across the capacit..." [Ans: N/A]

3.2 Parallel Plate Capacitance & Dielectric Slabs (High Yield — 9 PYQs)

Core Principle: Capacitance of parallel plate capacitor $C_0 = (\epsilon_0 A / d)$. With dielectric slab of constant K , $C = K C_0$. If battery disconnected: charge Q is constant, potential $V = V_0/K$, electric field $E = E_0/K$. If battery connected: V is constant, $Q = K Q_0$.

■ **Caution / JEE Pitfall:** Distinguish whether the battery is kept connected ($V = \text{const}$) or disconnected ($Q = \text{const}$) before inserting the dielectric slab.

■ **Speed Trick:** KCET Rule of Thumb: Disconnected battery implies Q stays constant. Connected battery implies V stays constant.

Representative Source PYQs:

- **[2026 Shift] Q#2 (MCQ):** "A parallel plate capacitor has a uniform electric field 'E' in the space between the plates. If the distance between the plates is 'd...' [Ans: C]
- **[2024 Shift] Q#3 (MCQ):** "A capacitor of capacitance $5\mu F$ is charged by a battery of emf $10 V$. At an instant of time, the potential difference across the capacit..." [Ans: N/A]
- **[2023 Shift] Q#1 (MCQ):** "A parallel plate capacitor of capacitance C_0 with a dielectric slab in between its plates is connected to a battery. It has a potential..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

In the circuit shown in the figure, the potential difference across the $4\mu F$ capacitor is

- (A) $3 V$
- (B) $4 V$
- (C) $9 V$
- (D) $12 V$

Correct Answer: None

Solution Analysis: The $9\mu F$ and $3\mu F$ capacitors are connected in parallel. In parallel combination, capacitances are added directly. $C_p = 9 + 3 = 12\mu F$ Now this $12\mu F$ equivalent capacitor is in series with the $4\mu F$ capacitor. For two capacitors in series, the equivalent capacitance is: $C_{eq} = (4 \times 12 / 4 + 12) = (48 / 16) = 3\mu F$ The battery voltage is $12V$. So the total charge supplied by the battery is: $Q = C_{eq} \times V = 3\mu F \times 12V = 36\mu C$ In a series combination, each capacitor has the same charge. So the charge

Benchmark Problem 2: KCET 2026 (Q#2)

A parallel plate capacitor has a uniform electric field 'E' in the space between the plates. If the distance between the plates is 'd' and area of each plate is 'A', the energy stored in the capacitor is

- (A) $(1/2)\epsilon_0 E^2$
- (B) $\epsilon_0 Ead$
- (C) $(1/2)\epsilon_0 E^2 Ad$ [Correct]
- (D) $\frac{E^2 Ad \epsilon_0}{2}$

Correct Answer: C

Solution Analysis: For a parallel plate capacitor, the energy stored is $U = (1/2)CV^2$ For a parallel plate capacitor in air or vacuum, $C = \epsilon_0 Ad$ Also, the electric field between the plates is uniform, so $E = (V/d)$ Hence, $V = Ed$ Substituting these in the energy formula, $U = (1/2)(\epsilon_0 Ad)(Ed)^2$ $U = (1/2)\epsilon_0 AdE^2 d$ $U = (1/2)\epsilon_0 E^2 Ad$ So, the energy stored in the capacitor is $(1/2)\epsilon_0 E^2 Ad$ Correct option: C.